

Forecast comparison & XAI report

Folder: artifacts/experiments/comparison/8a88d7fbc4d5

Executive summary

- Best (validation RMSE): **linear__91e98761**
- Best (test RMSE): **linear__91e98761**

Metrics (per run)

method	validation.rmse	validation.mae	validation.r2	validation.n	test.rmse	test.mae	test.r2	test.n
AR(1)__1543452a	1.0648	0.6798	-0.8639	84.0000	1.6208	0.8798	-0.1003	416.0000
AR(p auto)__9a2891c6	1.0124	0.6859	-0.6849	84.0000	1.6652	0.8770	-0.1615	416.0000
linear__91e98761	0.7573	0.5384	0.0572	84.0000	1.4125	0.7644	0.1643	416.0000

Error analysis (MAE + Diebold–Mariano)

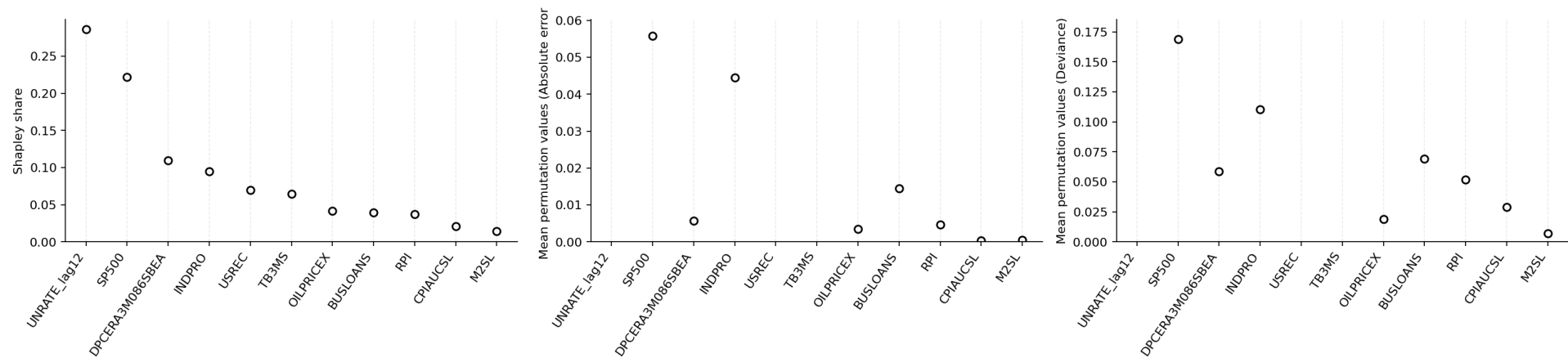
model	Ensemble	1990-1999	2000-08/2008	09/2008-11/2019
AR1	0.8798 (0.080)	0.4827 (0.421)	0.8019 (0.178)	0.9997 (0.533)
ARP	0.8770 (0.124)	0.4643 (0.626)	0.7946 (0.198)	0.9770 (0.674)
LINREG	0.7644	0.4360	0.6622	0.9206

Explainable AI (XAI)

Top features (SHAP share + permutation importance)

model	feature	shapley_share	mean_perm_abs_err	mean_perm_deviance	n_snapshots
LINREG	UNRATE_lag12	0.286	-0.0354	-0.1685	416
LINREG	SP500	0.222	0.0557	0.1686	416
LINREG	DPCERA3M086SBEA	0.109	0.0057	0.0587	416
LINREG	INDPRO	0.095	0.0444	0.1101	416
LINREG	USREC	0.070	-0.0259	-0.0938	416
LINREG	TB3MS	0.065	-0.0166	-0.0293	416
LINREG	OILPRICEX	0.042	0.0035	0.0190	416
LINREG	BUSLOANS	0.039	0.0144	0.0691	416
LINREG	RPI	0.037	0.0046	0.0520	416
LINREG	CPIAUCSL	0.021	0.0003	0.0288	416
LINREG	M2SL	0.014	0.0006	0.0070	416

Variable importance panels

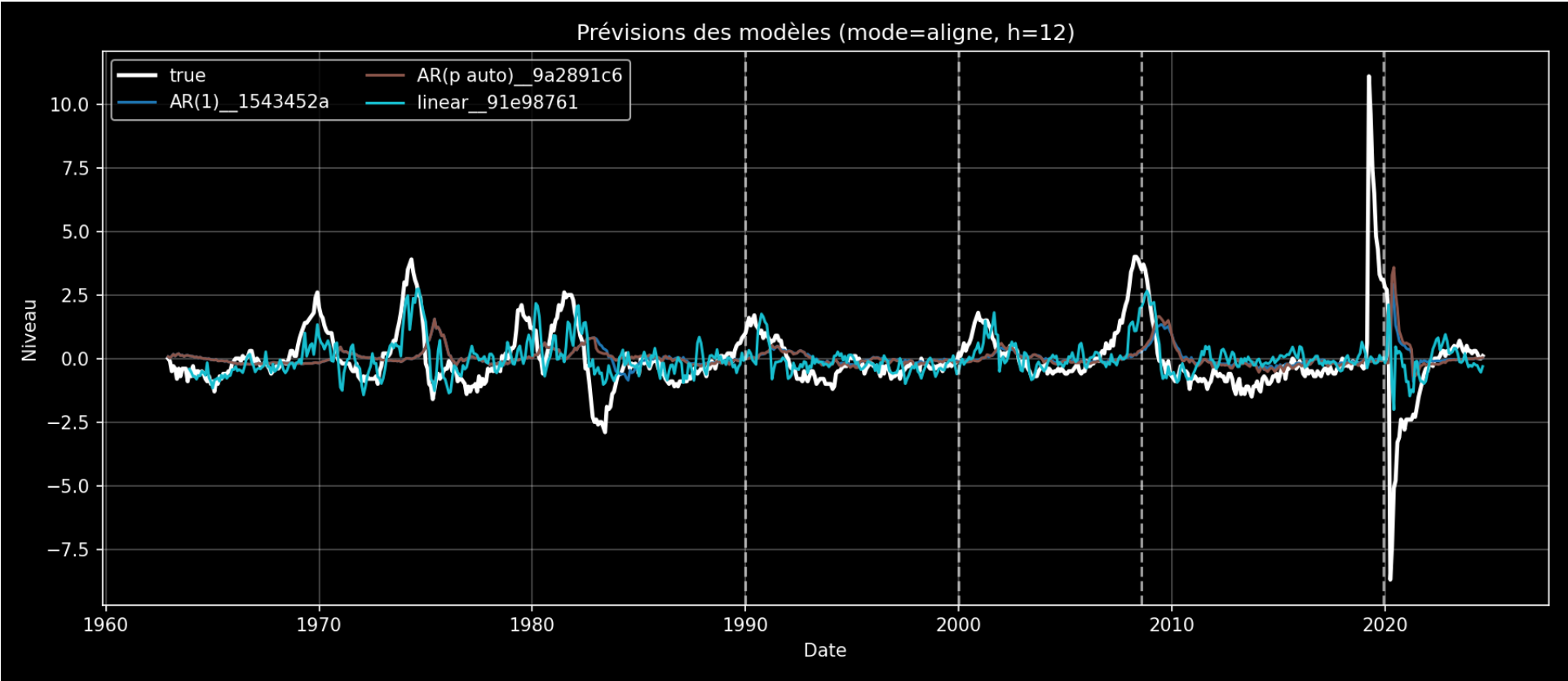


Statistical inference (Shapley regressions)

Forecasting	LINREG		
	β	p-value	Γ
UNRATE_lag12	-0.4116803238011639***	0.004	0.293
SP500	0.6326455380522208***	0.004	0.222
DPCERA3M086SBEA	1.1860110397031416*	0.053	0.102
INDPRO	2.075088471490567**	0.045	0.091
USREC	-2.136105994525133**	0.028	0.072
TB3MS	-0.6265109378880103	0.508	0.059
BUSLOANS	2.1080473191119378*	0.055	0.047
RPI	-0.3355871164686902	0.838	0.041
OILPRICEX	4.826880664058942**	0.012	0.038
CPIAUCSL	6.178557335845836	0.244	0.021
M2SL	9.731747970782536**	0.042	0.013

Figures

Global forecasts



Absolute errors

